Budget-Robust Station Selection for Microtransit Under Demand Uncertainty

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1 Introduction

Microtransit is a form of transit that seeks to fill the gap between existing services such as private-hire vehicles and scheduled public transit. It is an on-demand, publicly accessible shared-ride service. Like private-hire, it is flexible and on-demand, but it also benefits from the economies of scale of scheduled public transportation by using mid-size capacity vehicles and allowing for greater vehicle utilization.

However, having a fully flexible microtransit system is overly complex and reduces its aggregation capabilities, often having to make significant detours just to satisfy one or two customers. Thus, selection of these *virtual bus stops* (VBS) becomes an important strategic decision. The optimal selection enables better passenger aggregation, reducing total vehicle distance travelled and improving vehicle utilization, while offering passengers decreased waiting times and fewer detours than scheduled public transit.

While the VBS selection is able to adapt between service periods within the day (morning, afternoon, evening, night), the selection for each service period should be robust against shifts in the demand patterns. Within a service period, demand can fluctuate from day to day and the system should still provide a relatively similar experience across each day. We thus address this problem with a two-stage robust model where origin-destination (OD) demand varies within a budgeted uncertainty set.

2 Model

2.1 Nominal Model

Sets.

J set of candidate stations,

S set of service scenarios,

Ω set of origin-destination pairs,

A_{od} admissible pickup–dropoff station pairs for OD pair (o,d),

where

$$A_{od} = \left\{(j,k) \in J \times J : d^{\text{orig}}_{oj} \le W^{\max},\ d^{\text{dest}}_{dk} \le W^{\max}\right\}.$$

In the experimental instance, A_{od} is non-empty for all $(o,d) \in \Omega$.

Input parameters. The following parameters are assumed to be known:

$$
\begin{aligned}
q_{ods} &\geq 0 && \text{demand for OD pair } (o,d) \text{ in scenario } s,\\
d^{\text{orig}}_{oj} &\geq 0 && \text{walking distance from origin } o \text{ to station } j,\\
d^{\text{dest}}_{dk} &\geq 0 && \text{walking distance from station } k \text{ to destination } d,\\
c_{jk} &\geq 0 && \text{vehicle travel cost from station } j \text{ to station } k,\\
\lambda &\geq 0 && \text{weight on vehicle travel cost relative to walking distance},\\
W^{\max} &\geq 0 && \text{maximum walking distance allowed for a passenger},\\
l &\in \mathbb{Z}_+ && \text{number of stations to build},\\
k &\in \mathbb{Z}_+ && \text{number of stations to activate per scenario.}
\end{aligned}
$$

Decision variables. The model has three layers of binary decisions:

$$
y_j = \begin{cases} 1 & \text{station } j \text{ is built (long-run infrastructure decision)} \\ 0 & \text{otherwise} \end{cases}
$$

$$
z_{js} = \begin{cases} 1 & \text{station } j \text{ is activated and available for service in scenario } s \\ 0 & \text{otherwise} \end{cases}
$$

$$
x_{odjks} = \begin{cases} 1 & \text{OD pair } (o,d) \text{ is assigned to pickup station } j \text{ and dropoff station } k \text{ in scenario } s \\ 0 & \text{otherwise} \end{cases}
$$

Objective. The objective minimizes total demand-weighted travel cost across all scenarios:

$$
\min_{x,y,z} \sum_{s\in S} \sum_{(o,d)\in\Omega} \sum_{(j,k)\in A_{od}} q_{ods} \underbrace{\left(d^{\text{orig}}_{oj} + d^{\text{dest}}_{dk} + \lambda\, c_{jk}\right)}_{a_{odjks}} x_{odjks}. \tag{1}
$$

Constraints. The constraints enforce the two-stage structure:

$$
\begin{aligned}
&\sum_{(j,k)\in A_{od}} x_{odjks} = 1 && \forall (o,d), s && (2)\\
&x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} && \forall (o,d), (j,k) \in A_{od}, s && (3)\\
&z_{js} \leq y_j && \forall j, s && (4)\\
&\sum_j y_j = l, \quad \sum_j z_{js} = k && \forall s && (5)
\end{aligned}
$$

Constraint (2) requires every OD pair to be assigned to exactly one admissible station pair per scenario. Constraints (3) ensure that assignments can only use active stations. Constraint (4) requires that a station must be built before it can be activated. Constraint (5) fixes the number of built stations to l and the number of active stations per scenario to k.

2.2 Demand Uncertainty

2.2.1 Uncertainty Set

For each period s, demand lies in

$$
\mathcal{U}_s = \left\{ q_s : \underline{q}_{ods} \leq q_{ods} \leq \overline{q}_{ods} \ \forall (o,d), \ \sum_{(o,d)\in\Omega} q_{ods} \leq \overline{Q}_s \right\}. \tag{6}
$$

The per-OD bounds $[\underline{q}_{ods}, \overline{q}_{ods}]$ reflect the observation that demand for a given OD pair fluctuates around a historical range from day to day, but rarely departs dramatically from it. Setting $\underline{q}_{ods} = 0$ acknowledges that many OD pairs are absent on any given day, while $\overline{q}_{ods}$ is calibrated from empirical quantiles. The aggregate cap $\overline{Q}_s$ encodes the intuition that while demand can shift *across* OD pairs, the total number of trips in a time period is unlikely to surge simultaneously everywhere. The population served by the system does not change overnight. Together these two components avoid the excessive conservatism of a box uncertainty set, where every OD pair could independently attain its worst-case value at the same time. Since the bounds are integral, the inner maximization has an integral optimal extreme point, so integrality can be dropped without changing the worst-case value.

2.2.2 Robust Objective and Recourse

Given the uncertainty set (6), the robust objective protects against the worst-case demand realization in each scenario:

$$\min_{y,z} \sum_{s \in S} \max_{q_s \in \mathcal{U}_s} \Phi_s(z_s, q_s), \tag{7}$$

subject to (4)–(5). The recourse value $\Phi_s(z_s, q_s)$ is the optimal assignment cost for scenario s given fixed activated stations z_s and realized demand q_s:

$$\Phi_s(z_s, q_s) := \min_{x \geq 0} \sum_{(o,d) \in \Omega} \sum_{(j,k) \in A_{od}} q_{ods}\, a_{odjks}\, x_{odjks} \tag{8}$$

subject to (2)–(3). The assignment decision x is a *recourse* decision: it is made after demand is observed and adapts to the realized q_s. Since this inner problem decomposes by OD pair, letting $m_{ods}(z_s)$ denote the minimum assignment cost for pair (o, d) given z_s, the recourse value simplifies to

$$\Phi_s(z_s, q_s) = \sum_{(o,d) \in \Omega} q_{ods}\, m_{ods}(z_s). \tag{9}$$

This separability is key: it means the inner maximization over $q_s \in \mathcal{U}_s$ is a linear program in q_s for fixed z_s, which can be dualized to yield a tractable reformulation.

2.3 Derivation of the Robust Counterpart

Introducing epigraph variables η_s, the robust problem becomes

$$\min_{y,z,\eta} \sum_s \eta_s \quad \text{s.t.} \quad \eta_s \geq \max_{q_s \in \mathcal{U}_s} \Phi_s(z_s, q_s) \quad \forall s. \tag{10}$$

Substituting (9) and dualizing the inner LP gives dual variables $\alpha_s \geq 0$, $\beta_{ods} \geq 0$, $\gamma_{ods} \geq 0$ such that

$$\eta_s \geq \overline{Q}_s\, \alpha_s + \sum_{(o,d)} \overline{q}_{ods}\, \beta_{ods} - \sum_{(o,d)} \underline{q}_{ods}\, \gamma_{ods} \qquad \forall s, \tag{11}$$

$$\alpha_s + \beta_{ods} - \gamma_{ods} \geq m_{ods}(z_s) \qquad \forall (o, d),\ s. \tag{12}$$

Expanding $m_{ods}(z_s)$ with the assignment variables x and eliminating η_s (tight at optimality) yields the final MILP:

$$\min_{x,y,z,\alpha,\beta,\gamma} \sum_s \overline{Q}_s\, \alpha_s + \sum_s \sum_{(o,d)} \overline{q}_{ods}\, \beta_{ods} - \sum_s \sum_{(o,d)} \underline{q}_{ods}\, \gamma_{ods} \tag{13}$$

subject to

$$\alpha_s + \beta_{ods} - \gamma_{ods} \geq \sum_{(j,k)\in A_{od}} a_{odjks}\, x_{odjks} \qquad \forall (o,d),\ s, \tag{14}$$

$$\sum_{(j,k)\in A_{od}} x_{odjks} = 1 \qquad \forall (o,d),\ s, \tag{15}$$

$$x_{odjks} \leq z_{js}, \quad x_{odjks} \leq z_{ks} \qquad \forall (o,d),\ (j,k) \in A_{od},\ s, \tag{16}$$

$$z_{js} \leq y_j \qquad \forall j,\ s, \tag{17}$$

$$\sum_j y_j = l, \quad \sum_j z_{js} = k \qquad \forall s, \tag{18}$$

with $x_{odjks}, \alpha_s, \beta_{ods}, \gamma_{ods} \geq 0$ and $y_j, z_{js} \in \{0, 1\}$.

3 Experiments

We use operational data from a microtransit system in Zhuzhou, China. Table 1 describes the three dataset variants; the sparse 40-station instance is used for the main comparison because it preserves most of the order volume while substantially shrinking the calibrated OD support.

Table 1: Dataset variants for the April calibration and May backtest windows.

Metric	Full Zhuzhou	Sampled	Sparse
Stations	84	40	40
Directed segments	6,972	1,560	1,560
April orders	2,741	2,560	2,560
April ODs with orders	473	389	130
May orders	31,546	30,100	30,100
May ODs with orders	1,412	949	131

3.1 Experimental Design

We calibrate on April 1–3, 2025 and evaluate on May 1–31, 2025. This is an intentionally data-poor split that mimics operators who have not yet observed dense historical demand. We fix $l = 30$ built stations, a 300-second walking limit, $\lambda = 10$, and vary $k \in \{10, 15, 20\}$. For the robust model we vary the calibration quantile $\tau \in \{0.90, 0.95, 0.99\}$.

3.2 Uncertainty-Set Calibration

Scenarios correspond to four time-of-day periods. Lower bounds are fixed at $\underline{q}_{ods} = 0$; upper bounds $\overline{q}_{ods}$ are empirical quantiles of zero-padded daily OD counts; the cap $\overline{Q}_s$ uses the same quantile on total period demand. Table 2 shows that both 40-station variants produce large box-to-cap ratios. The ratio column reports $\left(\sum_{(o,d)} \overline{q}_{ods}\right)/\overline{Q}_s$, the ratio of the box worst case to the aggregate cap. Larger values indicate greater conservatism reduction from the cap. The sparse instance starts from much thinner OD coverage per period.

Table 2: Uncertainty-set diagnostics, 90th-percentile calibration.

Period	Sampled ODs	Sparse ODs	Sampled ratio	Sparse ratio
Morning	94	80	17.42	17.46
Afternoon	169	103	7.12	7.28
Evening	249	122	2.86	2.97
Night	120	87	13.16	13.31

4 Results

Both experiments evaluate on 30,100 May orders from the full-OD dataset (949 OD pairs). For the sparse experiment this is a distributional shift: models were calibrated on 130 OD pairs while the test set spans all 949.

4.1 Sampled-Network Results

With dense training OD coverage, all models achieve zero uncovered orders (Table 3). The nominal model has the lowest mean cost for every k; robust variants trade a small mean cost increase for lower variance.

Table 3: Sampled-network results, full-OD May orders. Bold = best in column within each k-block.

k	τ	Model	April mean cost	May mean cost	April std.	May std.	Uncovered
10	–	Nominal	**4266.09**	**4333.42**	1129.72	1116.39	**0**
10	0.90	Robust	4287.81	4359.76	1111.08	1102.67	**0**
10	0.95	Robust	4302.24	4392.66	**1108.89**	**1090.17**	**0**
10	0.99	Robust	4293.94	4369.71	1109.24	1098.48	**0**
15	–	Nominal	**3977.09**	**4049.94**	1145.30	1119.97	**0**
15	0.90	Robust	4018.31	4071.27	1106.48	1088.71	**0**
15	0.95	Robust	4023.38	4080.47	1105.51	1085.45	**0**
15	0.99	Robust	4030.38	4067.03	**1096.25**	**1066.90**	**0**
20	–	Nominal	**3924.95**	**3984.77**	1168.67	1142.49	**0**
20	0.90	Robust	3967.29	4003.44	**1116.51**	**1082.44**	**0**
20	0.95	Robust	3948.96	3978.06	1118.75	1090.00	**0**
20	0.99	Robust	3958.28	4005.45	1118.56	1091.22	**0**

4.2 Sparse Setting

When calibration data is thin, the nominal plan may not cover out-of-sample OD pairs. We simulate this by filtering calibration to the top 10% of OD pairs by demand frequency (389 $\rightarrow$ 130), giving the robust model a structural advantage.

4.3 Sparse-Network Results

The nominal model fits the sparse training data best but leaves 227 orders uncovered at $k = 10$ and 14 at $k = 15$. Robust models achieve complete coverage (Table 4). At $k = 20$ all models cover every order and costs converge.

Table 4: Sparse-network results, full-OD May orders. Bold = best in column within each k-block.

k	q	Model	April mean cost	May mean cost	April std.	May std.	Uncovered
10	–	Nominal	**4446.97**	**4305.57**	851.46	1092.41	227
10	0.90	Robust	4505.20	4363.35	845.97	1100.00	**0**
10	0.95	Robust	4521.04	4392.62	**836.33**	1094.39	**0**
10	0.99	Robust	4507.51	4368.66	840.04	**1089.52**	**0**
15	–	Nominal	**4182.48**	**4031.55**	881.21	1082.65	14
15	0.90	Robust	4234.00	4080.66	853.99	1059.06	**0**
15	0.95	Robust	4235.88	4089.67	**848.30**	**1058.01**	**0**
15	0.99	Robust	4236.58	4073.98	855.82	1067.54	**0**
20	–	Nominal	**4136.13**	**3972.53**	901.73	1094.37	**0**
20	0.90	Robust	4149.90	3985.01	886.97	1082.80	**0**
20	0.95	Robust	4155.78	3998.38	**883.00**	**1077.17**	**0**
20	0.99	Robust	4148.22	3984.41	887.77	1079.39	**0**

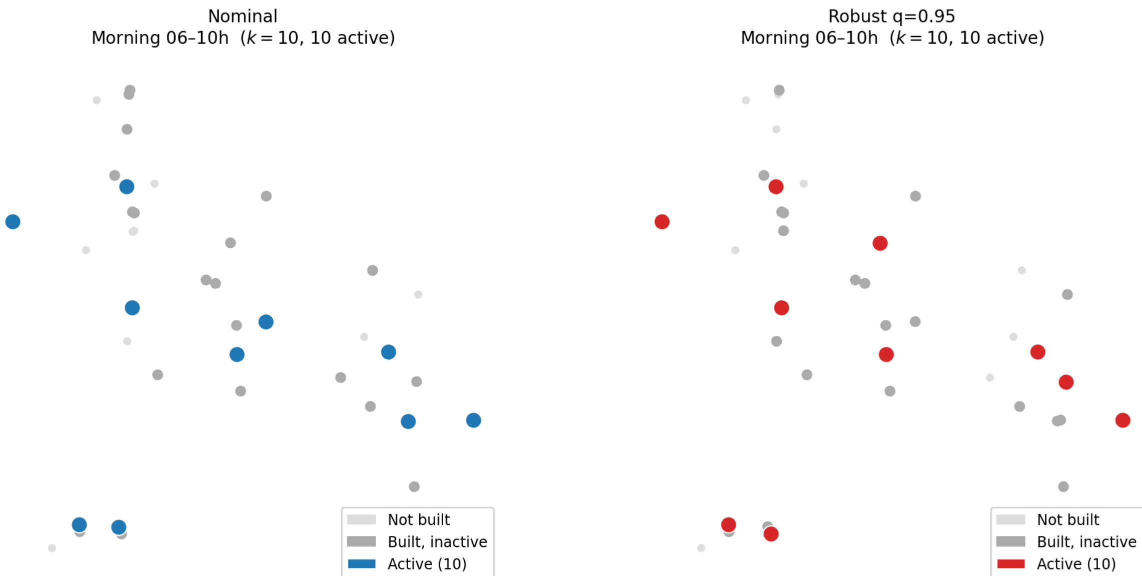
All three models build the same 30 stations (see Figure 3 in the appendix), so the divergence is entirely in which k stations are activated per period. Figure 1 shows the morning period at $k = 10$: the nominal model places all 10 activations in the high-demand core near the centre of the network, while the robust ($q = 0.95$) model distributes activations more broadly, including peripheral stations that are rarely observed in the sparse calibration data. This broader coverage is what eliminates the 227 uncovered orders.

However, for orders that *are* covered, the cost distributions are nearly identical. Figure 2 shows that CDFs for nominal and robust largely overlap, with robust variants shifted only slightly to the right in the upper tail. This reveals a gap between the expected and actual benefit of robustness: the model was designed to hedge against worst-case costs, but in practice the recourse function is simple enough that once a station is within reach, the assignment cost is similar regardless of which model built the plan. The main benefit in the sparse setting is therefore service availability, specifically ensuring that peripheral demand is not left uncovered, rather than cost reduction for served orders.

5 Conclusion

A budgeted uncertainty set is a practical way to model uncertain OD demand without extreme conservatism: per-OD bounds keep each pair within its historical range while the cap $\overline{Q}_s$ prevents simultaneous surges. When data is dense, robustness primarily reduces cost variance. When data is sparse, the nominal plan concentrates activations in the observed high-demand core and leaves peripheral orders uncovered, while the robust plan achieves full coverage at a modest cost premium. Coverage can alternatively be enforced via explicit constraints, which may be more efficient when variance reduction is not a goal.

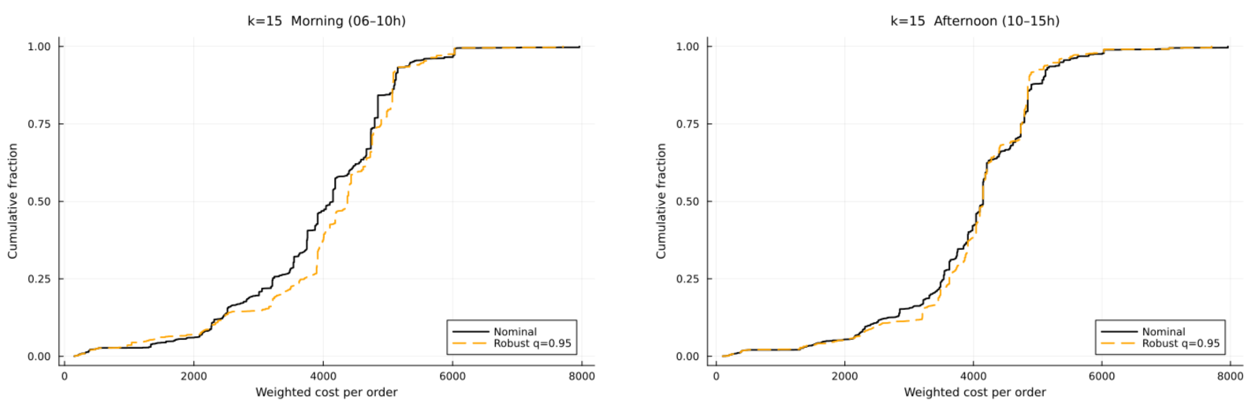
Future work could incorporate richer recourse functions reflecting true microtransit operations (vehicle capacity, rebalancing, multi-period dynamics), and identify which demand regimes, such as low-density or spatially shifting demand, most benefit from a robust plan.



(a) Nominal

(b) Robust ($q = 0.95$)

Figure 1: Morning-period activations for $k = 10$. Filled = active; hollow = built but inactive. Nominal clusters all activations in the high-demand core; robust spreads them across the network.



(a) Morning (06–10h)

(b) Afternoon (10–15h)

Figure 2: Cost CDFs for $k = 15$, sparse experiment, May full-OD test set. Nominal and robust distributions largely coincide for covered orders.

A Additional Figures

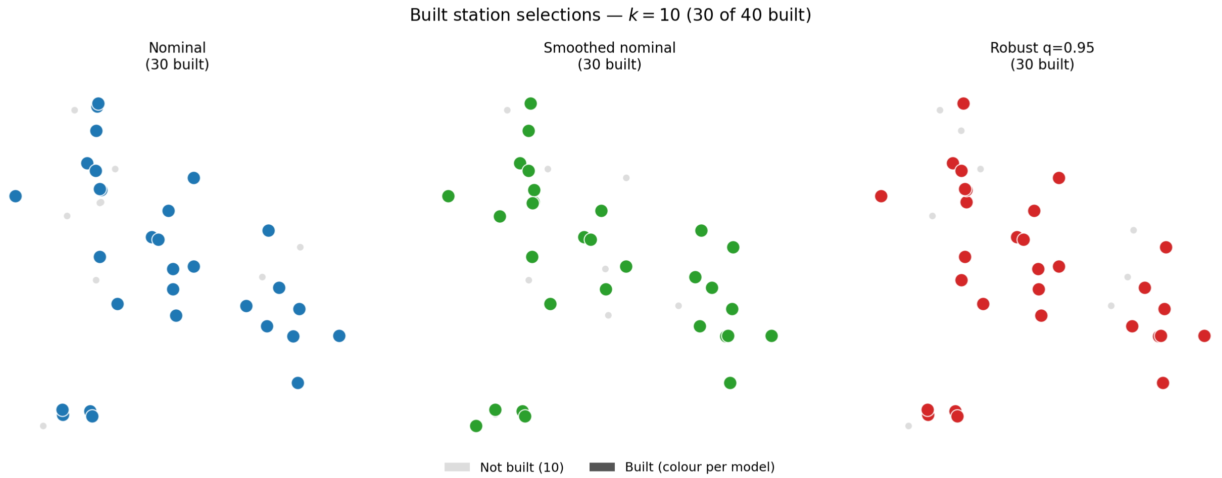


Figure 3: Built station selections for $k = 10$. Coloured dots = built (30 of 40); gray = not built. All three models select the same set of 30 stations.

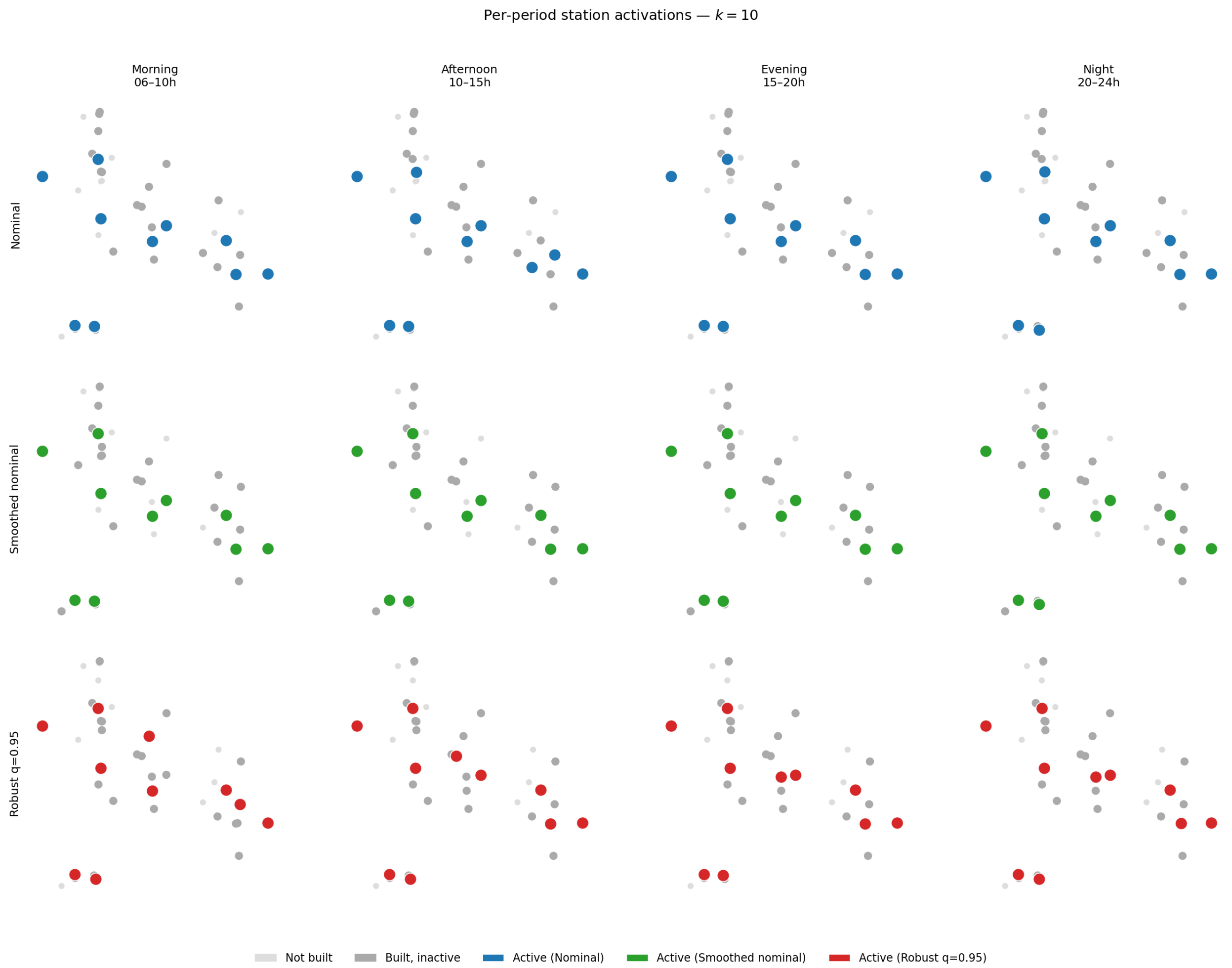
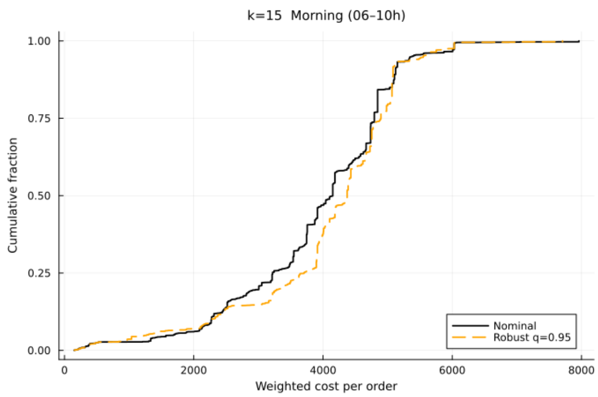
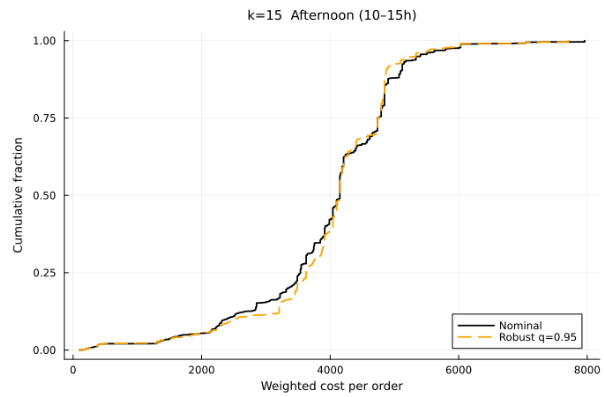


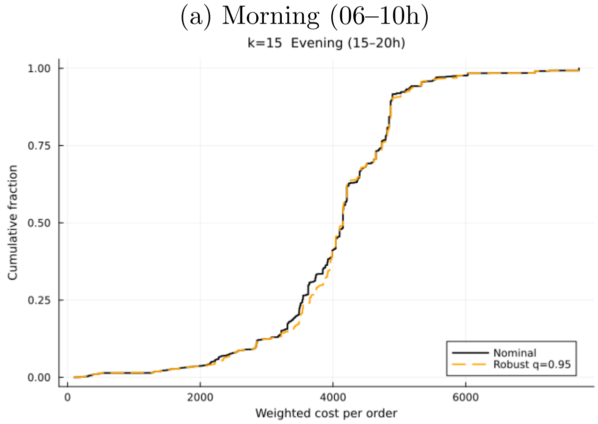
Figure 4: Per-period activations for $k = 10$ across all four time-of-day scenarios. Filled = active; hollow = built but inactive; gray = not built.



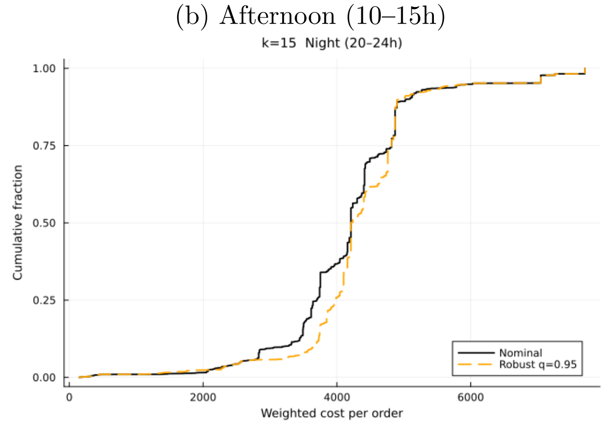
(a) Morning (06–10h)



(b) Afternoon (10–15h)



(c) Evening (15–20h)



(d) Night (20–24h)

Figure 5: Cost CDFs for $k = 15$ across all four time-of-day scenarios, sparse experiment, May full-OD test set.